

1

a

i

Since all $\forall i, 0 \leq p_i \leq 1$. This means for some $k : 1 \leq k < n$ then $\prod_{i=1}^k p_i \leq \prod_{i=1}^{k+1} p_i$

Thus in a machine system, we run into a problem if we have all p_i be nonzero but we have multiple p_i near 0 or at the points where representing the smallest possible nonzero number on our machine. Then we end up with a truncation error and likely rounding down to 0.

It is important to note that since multiplication is a benign operation there is no issue in errors exploding out but instead the issues arise for truncation errors.

ii

Let $P = \sum_{i=1}^n \ln p_i$, then $e^P = \prod_{i=1}^n p_i$

This is better because since each p_i is bound between 0 and 1, then $\ln p_i$ must be less than or equal to 0. Thus even though we are using addition operators the addition is benign since all $\ln p_i$ are the same sign (negative). It is important to note that we have issues for $p_i = 0$ since $\ln 0$ is undefined and approaches $-\infty$

iii

iv

b

i

Since x_i is not bound, then e^{x_i} can be beyond what we can represent on our machine is if $|x_i|$ is really large resulting in e^{x_i} being so large we cannot represent it or alternatively too small we can't represent it (depending on the sign of x_i). Note that if e^{x_i} is really small, then usually we do not care unless ALL x_i are small because then the denominator and numerator are really small. Instead the more dangerous case for this (and more likely) is a few x_i that are really large such that e^{x_i} is beyond what we are able to represent on the machine.

ii

We could instead change our exponents to prevent the top and bottom from exploding with the following:

$$\frac{e^{x_i}}{\sum_j e^{x_j}}$$

$$\frac{e^{-m} e^{x_i}}{e^{-m} \sum_j e^{x_j}}$$

Where $m = \max_i x_i$

$$\frac{e^{x_i - m}}{\sum_j e^{x_j - m}}$$

Since $m = \max_i x_i$ and we are finding the softmax, then

$$\frac{1}{\sum_j e^{x_j - m}}$$

iii

iv

2

a

$$x = 2x - yx^2$$

$$\rightarrow yx^2 + x = 2x$$

$$\rightarrow yx^2 = x$$

$$\rightarrow yx = 1$$

$$\rightarrow x = \frac{1}{y}$$

Or alternatively, $f(\frac{1}{y}) = \frac{2}{y} - \frac{y^2}{y} = \frac{2}{y} - \frac{1}{y} = \frac{1}{y}$

b

Relative error $\hat{e}_n = \left| \frac{\frac{1}{y} - x_n}{\frac{1}{y}} \right| = |1 - yx_n|$

And e_{n+1} will be $|1 - yx_{n+1}|$ so

$$\rightarrow \hat{e}_{n+1} = |1 - y(2x_n - yx_n^2)|$$

$$\rightarrow \hat{e}_{n+1} = |1 - 2yx_n - y^2x_n^2|$$

$$\rightarrow \hat{e}_{n+1} = |(1 - yx_n)^2|$$

$$\rightarrow \hat{e}_{n+1} = (e_n)^2$$

Therefore we only converge when our relative error e_n is less than 1 because squaring decreases for decimal points, stays same at 1 and goes to infinity at greater than 1.

Note that this is for both positive and negative values

Therefore if $|1 - yx| < 1$ then we will converge. Thus the range of starting x is $0 < x_0 < \frac{2}{y}$

c

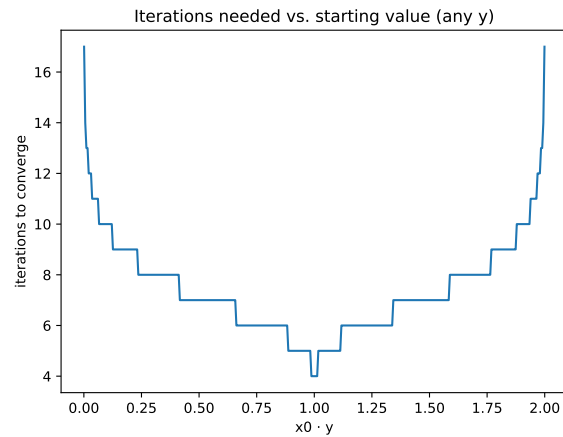
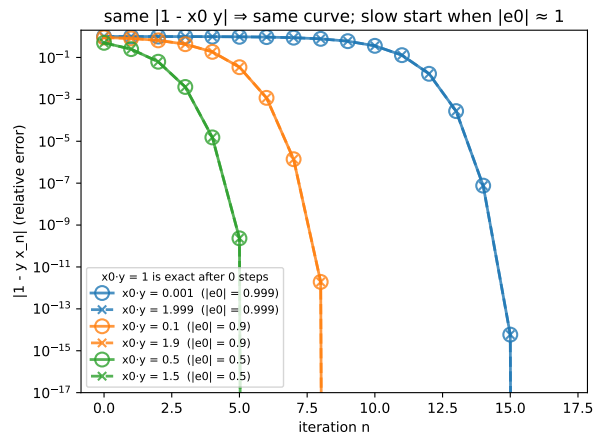
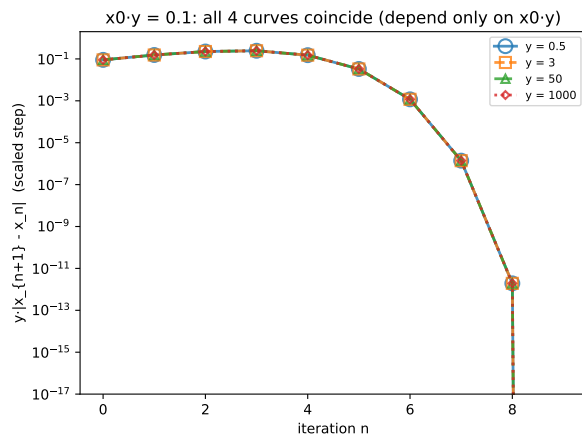
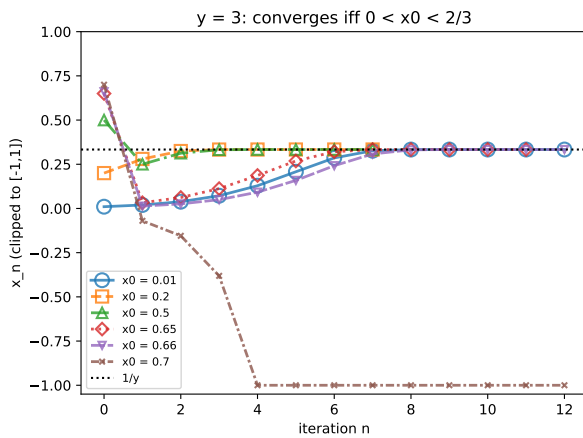
We have the exact case of quadratic convergence here because the relative error at each step is approximately the square of the relative error of the previous steps. So by the notes, the number of correct decimal digits doubles with each iteration.

d

y	x0	x0*y	iters	final x	1/y	result
3	0.01	0.03	12	3.333333e-01	3.333333e-01	converged
3	0.3	0.9	5	3.333333e-01	3.333333e-01	converged
3	0.6	1.8	9	3.333333e-01	3.333333e-01	converged
3	0.66	1.98	12	3.333333e-01	3.333333e-01	converged
3	0.6667	2	22	-4.678211e+181	3.333333e-01	DIVERGED
3	0.7	2.1	12	-1.167662e+169	3.333333e-01	DIVERGED
3	-0.1	-0.3	10	-1.588077e+116	3.333333e-01	DIVERGED
0.25	0.1	0.025	12	4.000000e+00	4.000000e+00	converged
0.25	7.9	1.98	12	4.000000e+00	4.000000e+00	converged
0.25	8.1	2.02	14	-2.003875e+176	4.000000e+00	DIVERGED
100	0.0001	0.01	13	1.000000e-02	1.000000e-02	converged
100	0.019	1.9	10	1.000000e-02	1.000000e-02	converged
100	0.021	2.1	12	-3.502986e+167	1.000000e-02	DIVERGED

$y = 3.0$, $x_0 = 0.05$

n	x_n	$ x_n - 1/y $	$ x_{n+1} - x_n $	e_{n+1} / e_n^2
0	0.05000000000000000	2.833e-01	4.250e-02	3.0000
1	0.09250000000000000	2.408e-01	6.683e-02	3.0000
2	0.15933125000000000	1.740e-01	8.317e-02	3.0000
3	0.2425031583203125	9.083e-02	6.608e-02	3.0000
4	0.3085829712546454	2.475e-02	2.291e-02	3.0000
5	0.3314955920642549	1.838e-03	1.828e-03	3.0000
6	0.3333232014544171	1.013e-05	1.013e-05	3.0000
7	0.333333330253684	3.080e-10	3.080e-10	585.2991
8	0.3333333333333334	5.551e-17	0.000e+00	18014398509481984.0000
9	0.3333333333333334	5.551e-17	nan	nan



kachow

3

a

$$x = x \frac{y}{2^x}$$

$$\rightarrow 1 = \frac{y}{2^x}$$

$$\rightarrow 2^x = y \text{ note i'll use this later in 3b}$$

$$\rightarrow x = \log_2 y$$

b

Suppose $x^* = \log_2 y$

$e_n = x_n - \log_2 y = x_n - x^*$ for absolute error and $\hat{e}_n = \frac{x_n}{x^*} - 1$ for relative error

$e_{n+1} = x_{n+1} - x^*$ for absolute error

$$\rightarrow e_{n+1} = \frac{x_n y}{2^{x_n}} - x^*$$

From 3a we know that $\frac{y}{2^{x^*}} = 1$

$$\rightarrow e_{n+1} = \frac{x_n y}{2^{x_n}} - \frac{x^* y}{2^{x^*}}$$

$$\rightarrow e_{n+1} = y \left(\frac{x_n}{2^{x_n}} - \frac{x^*}{2^{x^*}} \right)$$

$$\rightarrow e_{n+1} = y \left(\frac{2^{x^*} x_n}{2^{x^*} 2^{x_n}} - \frac{2^{x^*} x^*}{2^{x^*} 2^{x_n}} \right)$$

$$\rightarrow e_{n+1} = y \left(\frac{2^{x^*} x_n - 2^{x^*} x^*}{2^{x^*} 2^{x_n}} \right)$$

From 3a we know that $y = 2^{x^*}$

$$\rightarrow e_{n+1} = \frac{2^{x^*} x_n - 2^{x^*} x^*}{2^{x_n}}$$

$$\rightarrow e_{n+1} = \frac{2^{x^*} x_n}{2^{x_n}} - x^*$$

$$\rightarrow e_{n+1} = 2^{x^* - x_n} x_n - x^*$$

$$\rightarrow e_{n+1} = 2^{-e_n} x_n - x^*$$

Thus relative error is now: $\hat{e}_{n+1} = \frac{2^{-e_n} x_n - x^*}{x^*}$

$$\rightarrow \hat{e}_{n+1} = \frac{2^{-e_n} x_n}{x^*} - 1$$

$$\rightarrow \hat{e}_{n+1} = \frac{x_n}{x^*} 2^{-\hat{e}_n x^*} - 1$$

$$x_n = x^* + e_n$$

$$\begin{aligned}
\rightarrow \hat{e}_{n+1} &= \frac{x^* + e_n}{x^*} 2^{-\hat{e}_n x^*} - 1 \\
\rightarrow \hat{e}_{n+1} &= (1 + \frac{e_n}{x^*}) x^* 2^{-\hat{e}_n x^*} - 1 \\
\rightarrow \hat{e}_{n+1} &= (1 + \hat{e}_n) 2^{-\hat{e}_n x^*} - 1 \\
\rightarrow \hat{e}_{n+1} &= (1 + \hat{e}_n) y^{-\hat{e}_n} - 1 \\
\rightarrow \hat{e}_{n+1} &= (1 + \hat{e}_n) e^{-\hat{e}_n \ln y} - 1 \\
\rightarrow \hat{e}_{n+1} &\approx (1 + \hat{e}_n)(1 - \hat{e}_n \ln y) - 1 \\
\rightarrow &= 1 - \hat{e}_n^2 \ln y + \hat{e}_n - \hat{e}_n \ln y - 1 \\
\rightarrow &= -\hat{e}_n^2 \ln y + \hat{e}_n(1 - \ln y) \\
\rightarrow &\approx \hat{e}_n(1 - \ln y)
\end{aligned}$$

Thus linear convergence for $|1 - \ln y| < 1$ because then $|e_n| > |e_{n+1}|$ when $|e_n| > 0$

If we solve this then actually convergence for $1 < y < e^2$

c

From 3.b, we have that we converge for values of y between 1 and e^2 so converges for values of x^* between 0 and $2 \log_2 e > 1$

This is from the lipschitz convergence for $|e_{n+1}| = \rho |e_n|$ converging for $|\rho| < 1$

Given $g(x) = x \frac{y}{2^x}$, then $g'(x) = \frac{y}{2^x}(1 - x \ln 2)$

Since $x \in (x^*, 1)$, $0 < \frac{y}{2^x} \leq \frac{y}{2^{x^*}} = 1$ and $0 < 1 - x \ln 2 < 1 - x^* \ln 2 < 1$

Thus $g'(x)$ is between 0 and 1 for the whole interval of $[x^*, 1]$ therefore we converge

d

$$0 < x^* < 1 \rightarrow 0 < \log_2 y < 1 \rightarrow 1 < y < 2$$

For decomposing y not in this range, since we are solving for $\log_2 y$ and this works for y between 1 and 2, we can generalize this into $k + \log_2 m$ where m is now bound between 1 and 2.

What this means for y is $\log_2 y = k + \log_2 m \rightarrow y = m 2^k$ which can represent all real numbers greater than 0. Essentially multiplying or dividing y until the result is an m between 1 and 2 and some integer k

We know this algorithm converges because we know it converges for values of y between 1 and 2 and we can easily add /subtract back until we get our original desired number

e

$$f(x) = 2^x - y, f'(x) = 2^x \ln 2$$

$$\text{Thus } x_{n+1} = x_n - \frac{f(x)}{f'(x)} = x_n - \frac{2^x - y}{2^x \ln 2}$$

f